

Multiple Choice (10 marks)

1. Which of the following statements is not true?
 - (a) Humans cannot convert any fatty acids into docosahexaenoic acid
 - (b) Flaxseeds are a rich source of docosahexaenoic acid
 - (c) Docosahexaenoic acid levels are consistently higher in vegans than in omnivores
 - (d) Vegan diets provide an abundance of docosahexaenoic acid naturally
 - (e) Lack of docosahexaenoic acid in vegans causes visual and cognitive impairment
2. How long can a cannula remain in situ?
 - (a) 96 hours.
 - (b) 36 hours.
 - (c) 60 hours.
 - (d) 2 weeks.
 - (e) 1 week.
3. Which are the FIVE main series of apoproteins that have been identified?
 - (a) apoA, apo(a), apoB, apoC and apoH
 - (b) apoA, apoB, apoC, apo E, and apoL
 - (c) apoA, apo(a), apoB, apoC and apoE
 - (d) apoA, apoB, apoC, apoD, and apoF
 - (e) apoA, apoB, apoC, apo(a) and apoM
4. In a condition such as schizophrenia the recurrence risk will be greatest for which of the following relatives of an affected individual?
 - (a) Aunt
 - (b) Nephew
 - (c) Brother
 - (d) Grandchild
 - (e) Great-grandchild
5. Muscle lactate production increases when:
 - (a) the muscle is fully hydrated.
 - (b) oxygen levels in the muscle are low.
 - (c) glucose levels in the muscle are low.
 - (d) oxygen is readily available.
 - (e) the pH of the muscle falls.

True / False (10 marks)

Answer True or False

6. True or False: Biotin plays a crucial role in the carboxylation of pyruvate to form oxaloacetate during metabolic processes.
7. True or False: The total daily dose of Cotrimoxazole prescribed at 960 mg every 12 hours is 1.92 grams.
8. True or False: Inspecting the nail-bed curvature from below is the most reliable method to assess for clubbing.
9. True or False: Grandparents feel most appreciated when their college-age grandchildren regularly send them daily text messages.
10. True or False: Dental plaque is most likely to accumulate on the palatal surfaces of the upper molars and the labial surfaces of the lower incisors.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between optimism and longevity in older adults, analyzing how pessimism may influence health outcomes and lifespan.
12. Discuss the role of amino acids in gluconeogenesis, specifically analyzing why tryptophan cannot serve as a substrate in this metabolic pathway.

End of Paper